Arav Palsule

Thane, Maharashtra — 9833067265 — palsulearav934@gmail.com
LinkedIn — GitHub — Portfolio

B.E. Data Science (2027) — CGPA: 8.34/10

Professional Summary

Aspiring AI/ML Engineer with hands-on experience in Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, and Generative AI. Skilled in Python, SQL, data preprocessing, feature engineering, predictive modeling, and model evaluation. Experienced in building LLM-powered applications, Retrieval-Augmented Generation (RAG) systems, multi-agent workflows using LangGraph, and AI automation solutions. Familiar with context engineering, prompt engineering, embeddings, and API integrations.

Education

A. P. Shah Institute of Technology, Thane
Bachelor of Engineering in Data Science — 2023 – 2027
CGPA: 8.34/10

Technical Skills

Programming Languages: Python, SQL, JavaScript
Machine Learning: Supervised Learning, Predictive Modeling, Feature Engineering, Data Preprocessing, Model Training, Model Evaluation, Statistical Analysis
Artificial Intelligence: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Transfer Learning, Embeddings
Generative AI and Agentic AI: Large Language Models (LLMs), RAG, Vector Search, AI Agents, Multi-Agent Systems, LangGraph, Prompt Engineering, Context Engineering, Claude Skills
Libraries and Frameworks: Pandas, NumPy, Matplotlib, Seaborn, Flask
Databases and APIs: MySQL, SQLite, Supabase, REST APIs, API Integration
Tools and Platforms: Git, GitHub, Zapier, Zoho

Projects

Radiator Routes – AI-Powered Travel Planning Platform

- Built a **five-model AI travel workflow**: Jenny chatbot, itinerary planner, plan verifier, dynamic replanner, and Travel Memory.
- Used **Multi-Agent Orchestration** with **LangGraph** to collect group size, dates, budget, and interests, then generate personalized itineraries.
- Implemented **14 deterministic validation checks** for budget, scheduling, travel feasibility, geographic validity, and opening hours, with an LLM-based repair flow.
- Added disruption-aware replanning and **Least Misery** scoring to adapt itineraries and balance group preferences.

Technologies Used: Python, React, TypeScript, Vite, LangGraph, Supabase, Groq, WebLLM

CrackX – AI-Powered Road Damage Detection System

- Integrated **YOLOv8** to classify and localize cracks and potholes from road photos and videos, with severity assessment.
- Connected a **five-role workflow** for citizens, Road Safety Officers, contractors, compliance officers, and administrators.
- Implemented contractor assignments, before-and-after photo evidence, and a **two-step approval** flow before citizen ratings and feedback.
- Added duplicate-report checks and time-range reports for monitoring repair activity.

Technologies Used: Python, FastAPI, React Native, Expo, TypeScript, YOLOv8, OpenCV, Supabase

Dengue Outbreak Prediction – Machine Learning

- Built a forecasting pipeline using historical dengue cases, weather, geographic attributes, and temporal features.
- Compared **four regression algorithms**: Random Forest, Extra Trees, HistGradientBoosting, and Ridge Regression.
- Evaluated models using chronological holdout and rolling-window validation with **MAE, RMSE, and SMAPE**.

Technologies Used: Python, Pandas, NumPy, Scikit-learn, Random Forest, Extra Trees, HistGradientBoosting, Ridge Regression

CineRec AI – AI-Powered Video Highlight Generator

- Developed a **five-stage video pipeline**: scene extraction, emotion classification, importance ranking, semantic music matching, and reel composition.
- Applied **DistilBERT** for scene-emotion classification and embedding-based retrieval for semantic music matching.

- Automated timestamps, clip extraction, soundtrack alignment, and final video rendering.

Technologies Used: Python, FastAPI, Celery, Redis, PyTorch, DistilBERT, FFmpeg, MoviePy, Supabase

Experience

AI Intern – Ennoble Social Innovation September 2026 – October 2026

- Created custom Claude Skills using structured context engineering to automate business tasks, including PDF formatting according to internal document structures and standards.
- Developed an employee focus-monitoring prototype integrating face detection and emotion analysis to estimate focus scores and identify potential sleep or inactivity events.
- Designed automated screenshot capture and HR reporting workflows to communicate detected events and employee focus patterns.
- Built AI-powered business process automations using Zapier, LLMs, and API integrations to streamline repetitive operational tasks.

Achievements

- **Rank 10** – C.R. Rao Statistics & Probability Olympiad: Ranked among 2.5 lakh+ participants across Asia.
- **AIR 3192, AICR 35** – Aryabhatta National Maths Competition 2026: Scored 100% in Round 1 and 80% in Round 2.
- **2nd Runner-Up** – QUASAR 4.0 Hackathon: Delivered a technical solution under a 24-hour development deadline.
- **Elite Rank** – NPTEL Multivariable Calculus: Recognized for mathematical and quantitative problem-solving skills.

Leadership

Founding President – Maths & Applied Science Club — 2026

- Founded and led the club, organizing 25+ technical events to promote mathematics, science, and technical learning among students.

Research Member – DSA Club — 2025 – 2026

- Explored LLMs, Retrieval-Augmented Generation (RAG), and machine learning pipelines; mentored junior members in technical concepts.